

SAHIL MOURYA

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EXPERIENCE

Dolat Capital

Software Developer Engineer

Mumbai, India

June 2023 - Present

- Built a **low-latency Order Management System (OMS) in C++** to manage and update order states based on real-time exchange responses.
- Developed a **low latency C++ Adapter class** for live exchange connectivity, processing strategy signals to construct and transmit order messages and handle confirmations—enabling low latency end-to-end order placement in **nanoseconds** with **99.9% system availability**.
- Engineered a data-forwarding solution enabling cross-protocol message routing via **TCP, UDP, and RTI** achieving optimized resource utilization and scalable multi-module integration.
- Designed and implemented new packet structures and constructors for **FIX protocol** communication with the MCX exchange API, enhancing data exchange efficiency and seamless system integration.
- Optimized the Real-Time Market Feed **Qt UI** for Equity Products by improving data processing and resource management, cutting window loading times by **50%** and ensuring seamless performance.
- Developed a robust low latency Middleware module to streamline bidirectional message handling between the UI and backend, utilizing **multithreading, socket programming, and TCP Protocol**, achieving a **25%** improvement in CPU utilization and system performance.
- Utilized **RTI (Real-Time Innovations)** to handle high-volume data streams from publishers, introduced new packet types, and leveraged multithreading to ensure real-time UI updates, boosting data processing efficiency by **30%**.
- Developed and implemented a Trading UI feature enabling concurrent message transmission to multiple servers using the **TCP Protocol**, improving system communication efficiency and scalability.
- Automated system workflows using efficient and robust **Bash scripts**, reducing manual effort by **70%** and errors by **90%** significantly improving process reliability and operational efficiency.

EDUCATION

Maulana Azad National Institute of Technology

B.Tech , Electronics And Communication

Cumulative Grade: 7.8/10

Bhopal, MP

July 2019 - May 2023

Yash Public School

Senior Secondary Examination

Percentage: 93.6%

Nagda, MP

April 2018 - March 2019

SKILLS

Programming Languages: C++, C, Bash, Python, Java, PL/pgSQL

Technologies: Qt with C++

Databases: PostgreSQL, MySQL

Technical Skills: Socket Programming, Multithreading

Other Skills: Operating system, DBMS, Linux, Data Structures and Algorithm, Eclipse IDE, Git

PROJECTS

Low-Latency Order Matching Engine [GitHub](#)

- Designed and implemented a low-latency multithreaded order matching engine in C++17 using priority queues, custom comparators, STL containers, and RAII-based synchronization with mutex and lock_guard.
- Integrated order lifecycle management (place/cancel), signal handling, and asynchronous logging with file I/O, simulating exchange-grade matching logic (price-time priority).

CERTIFICATIONS AND CODING PROFILES

- [Excellence in Data Structures in C++](#) from Coding Ninjas
- [Excellence in C++](#) from Coding Ninjas
- Coding Profiles: [LeetCode \(sahil_mourya\)](#), (Solved over **500+ DSA problems** across platforms).